

HMS RESEARCH COMPUTING

Getting Started with O2

The HMS high-performance computing cluster,
and how to use it from your browser

45 minutes · no prior HPC experience needed

rchelp@hms.harvard.edu

Before we start

1

Ask questions whenever they come up

Unmute, raise a hand, or type in chat. Questions mid-slide are welcome.

2

No O2 account yet? That's fine

Training accounts are available for the class

3

Slides are yours to keep

github.com/hmsrc/user-training

Two hands-on exercises

Both run entirely in a web browser — nothing to install, no terminal required.

Have Chrome open and ready:

<https://o2portal.rc.hms.harvard.edu>

Please set your Zoom display name to match your name so we can mark attendance.

GOALS

By the end of these 45 minutes



Know what O2 is — and when you need it

And when your laptop is the better tool.



Move your data on and off the cluster

Upload, download, and know which filesystem to put it on.



Run a real analysis in a browser

RStudio, Jupyter, MATLAB or a desktop, on cluster hardware.



Read and write a job script

Enough Slurm to submit, monitor, and cancel your own jobs.

Agenda

1	O2 in ten minutes	What a cluster is, accounts, data security	8 min
2	Where your data lives	Home, group folders, and scratch	5 min
3	The O2 Portal	Files, apps, and a terminal in your browser	15 min
4	Jobs and the scheduler	The four decisions behind every job	7 min
5	Software on O2	Modules, R, Python, containers	4 min
6	Questions and where to get help		6 min



1

O2 in ten minutes

What it is, how to get on it, and what you may put on it

What is O2?

A shared Linux computer cluster, supported by HMS Research Computing, that you use from your own computer. Named for being the successor to the “Orchestra” cluster.

~13,000

CPU cores

across ~390 compute nodes

250 GiB

memory per job

up to 1 TiB on highmem nodes

240+

GPUs

most reserved for on-Quad labs

Slurm

job scheduler

on Red Hat Enterprise Linux

Two ways in. The **O2 Portal** in a web browser, or `ssh` from a terminal. Same cluster, same files, same scheduler — we start with the Portal.

Your laptop vs. a cluster

Stay on your laptop

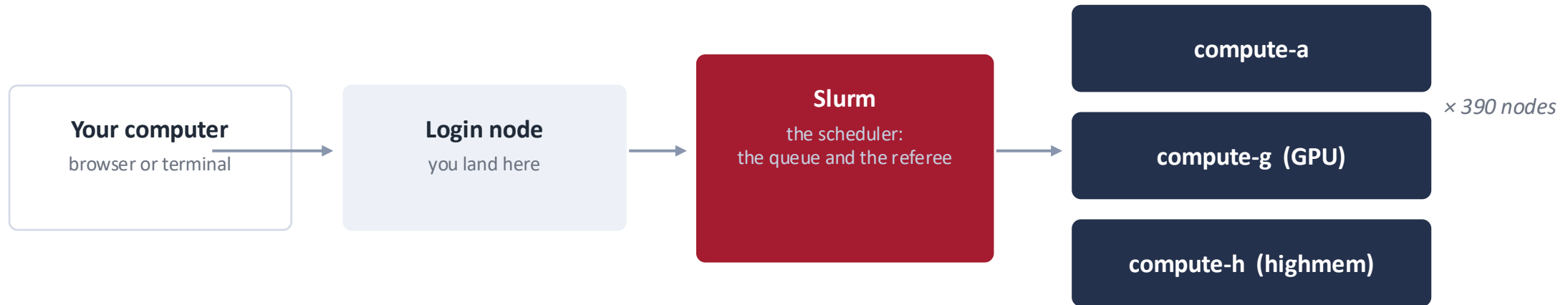
- The work finishes in a few minutes
- Your data fits comfortably in memory
- You are exploring, plotting, writing
- You need software requiring admin privileges

Move to O2

- A run takes hours or days
- You need far more memory than you have
- You need to run the same analysis over 50 samples
- You need a GPU, or licensed software like MATLAB
- Your data is too big for your disk

O2 is not faster at any one thing than a good laptop. It is bigger, and it will run a hundred things at once while you sleep.

How the O2 cluster works



Shared storage — /home · /n/groups · /n/data1 · /n/data2 · /n/scratch — visible from every node

The one rule that matters today. You log in to a login node, but you never compute on one. Ask Slurm for resources, and it hands you a compute node. Heavy work on a login node gets throttled and slows the cluster down for everyone — the Portal does this correctly for you automatically.

Getting an O2 account

You need a sponsoring PI and an HMS Account (HMS ID). Your O2 username and password are your HMS ID and password — there is nothing new to remember.

On-Quad

PI holds a primary or secondary appointment in an HMS Blavatnik Institute Basic or Social Science department

- Request at <http://rc.hms.harvard.edu/o2user>
- No charge for compute or storage

Off-Quad

Affiliate hospitals (MGH, BWH, BCH, BIDMC, DFCI) and other Harvard schools

- Register with the RC Core in PPMS first
- Billed quarterly for jobs and group storage
- /home and /n/scratch remain free

Two-factor authentication is required for every login. Set up Duo on your phone before your first login.

For more information, check “[How to request or retain an O2 account](#)” on the O2 wiki.

Before you upload anything

O2 is for Harvard data security Level 3 and below.

No standard O2 filesystem is encrypted. Do not place HIPAA-protected or other Level 4+ data on O2 unless it has been de-identified first.



Not sure what level your data is?

Ask your PI or department first, and check the [Harvard data security levels](#) before you transfer anything.



Sharing a folder does not change this

Group folders on /n/data1 and /n/groups carry the same restriction.

Where to read the policy

[“Security on O2”](#) on the O2 wiki

Visit the datamanagement.hms.harvard.edu website for resources and guidance in research data management

2

Where your data lives

Three filesystems, three different jobs

Storage at a glance

Location	What it's for	Size	Backed up?
/home/<hms_id>	Your scripts, small files, personal config	100 GiB, fixed	Yes — snapshots to 60 days
/n/groups · /n/data1 · /n/data2	Your lab's shared active research data	Expandable, by request	Yes — snapshots to 60 days
/n/scratch	Temporary and intermediate files while a job runs	25 TiB per user	No. None. Ever.
/n/files (research.files)	Collaboration space; transfer partition only		

Rule of thumb: keep the version of record in /home or a group folder, and let scratch hold anything you could regenerate.

You cannot compute against /n/files. Copy data from /n/files onto O2 storage first, using the transfer partition or the Portal. Ask rchelp for transfer access.

Fast, enormous, and temporary

1

If needed, create your own directory first

Run `/n/cluster/bin/scratch_create_directory.sh` from a login node.

2

25 TiB and 2.5 million files per user

Group directories under `/n/scratch/groups` still count against your personal quota.

3

Files not modified for 45 days are deleted

Automatically, with no way to get them back. Touching timestamps to dodge the purge is a policy violation.

4

There are no backups and no snapshots

If it only exists on scratch, it does not really exist.

Your scratch directory

```
/n/scratch/users/<first char>/<hms_id>
```

```
# e.g.   mfk8
```

```
/n/scratch/users/m/mfk8
```

The workflow that works

- Read your input from home or group storage
- Write intermediates to scratch
- Copy final results back off scratch

How much space am I using?

Jobs fail in confusing ways when a filesystem is full. Check before a big run, not after.

From a terminal

```
$ quota-v2

# usage for every filesystem
# you have access to.
# Refreshes hourly.
```

From the Portal

No commands needed — the Portal shows your quota usage on the dashboard and under the Files menu.

See “[Quota Usage on O2Portal](#)” on the wiki.

Two things worth knowing. The old `quota` command was retired in 2023 — use **`quota-v2`**. And to see what is eating a single folder, run `du --apparent-size -hs <dir>` — but do it on a compute node, not a login node.



3

The 02 Portal

Everything you just learned about — in a web browser

The Portal in one sentence

O2, in a browser tab. No ssh, no client to install, no X11 — you log in with your HMS ID and you are on the cluster.

1

Powered by Open OnDemand

An NSF-funded open-source project from the Ohio Supercomputer Center, used at HPC centres worldwide. HMS runs its own instance.

2

Nothing is simulated

The Portal is a front end. Your files are the real files and your jobs are real Slurm jobs, subject to the same partitions, limits and billing.

3

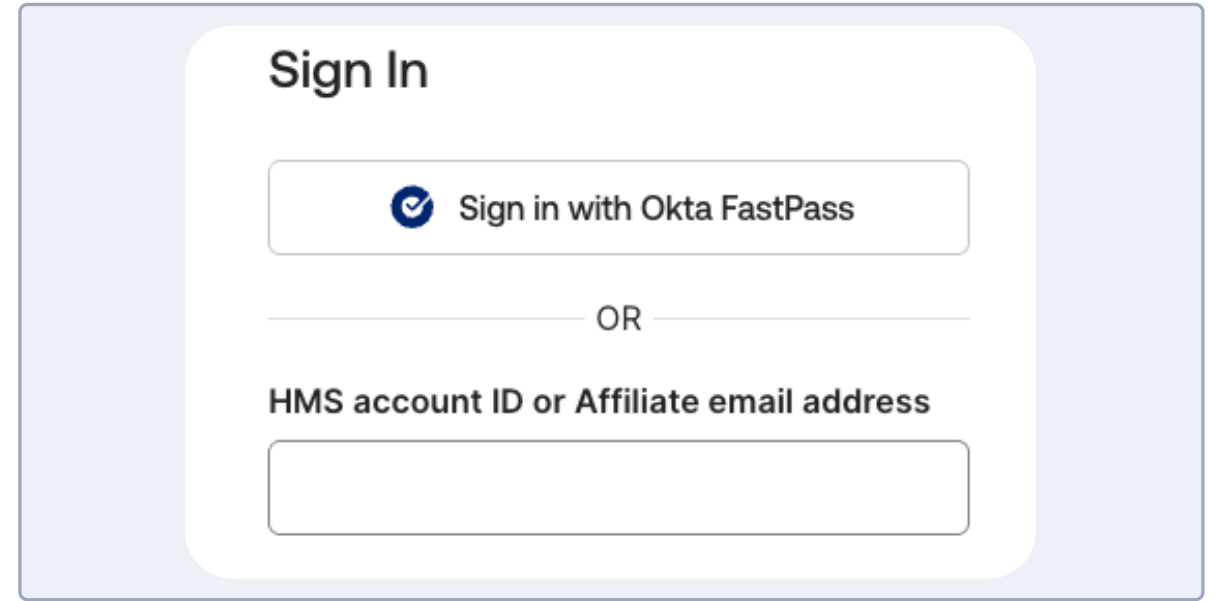
It survives a closed laptop

Sessions keep running if your browser disconnects. Start an analysis at work, reconnect to the same session from home.

`o2portal.rc.hms.harvard.edu`

Logging in takes three steps

- 1 Go to o2portal.rc.hms.harvard.edu**
In Chrome. You need an active O2 account — the Portal cannot create one for you.
- 2 Sign in with your HMS ID and password**
The same credentials you use for other HMS services.
- 3 Approve the two-factor prompt**
Okta push or fast-pass. Then you land on the Portal dashboard.

A screenshot of a web page titled "Sign In". It features a light blue background with a white rounded rectangle in the center. Inside the rectangle, the title "Sign In" is at the top. Below it is a button with a blue checkmark icon and the text "Sign in with Okta FastPass". Underneath the button is a horizontal line with the word "OR" in the center. Below the line is the text "HMS account ID or Affiliate email address" followed by an empty rectangular input field.

Sign In

 Sign in with Okta FastPass

OR

HMS account ID or Affiliate email address

Use Chrome, not Safari.

Some Open OnDemand apps misbehave in Safari. And if the Portal will not connect or a page hangs, clearing your browser cache fixes it far more often than anything else.

What you can do in the Portal



Files

Browse, upload, download, copy, move, rename



View / edit files

Open a script in a browser editor and save it



Terminal

A full O2 shell, no client needed



Submit jobs

Fill in a form instead of writing sbatch



Monitor jobs

Status, resources used, and the log files



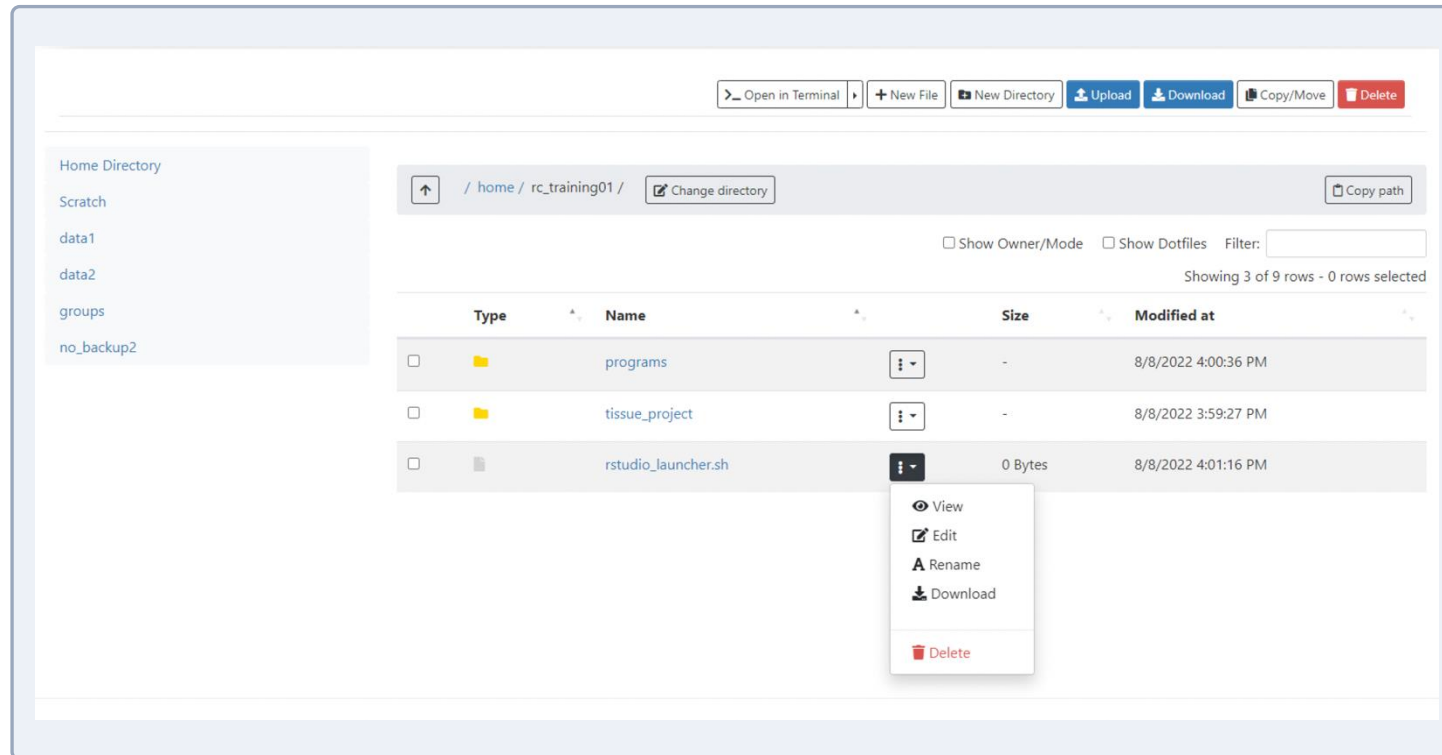
Applications

RStudio, Jupyter, MATLAB, VS Code, Desktop, RELION

Open OnDemand Files ▾ Jobs ▾ Clusters ▾ HMS-RC Applications ▾  My Interactive Sessions

 Help ▾  Logged in as User  Log Out

Browse, view, and edit — without a terminal



1

Files › Home Directory

Then use the file tree to reach /n/groups, /n/data1, or your scratch folder.

2

Edit opens a real editor

Syntax highlighting, save in place. Good for fixing a job script in ten seconds.

3

New File / New Directory

Replaces nano, mkdir and touch for everyday work.

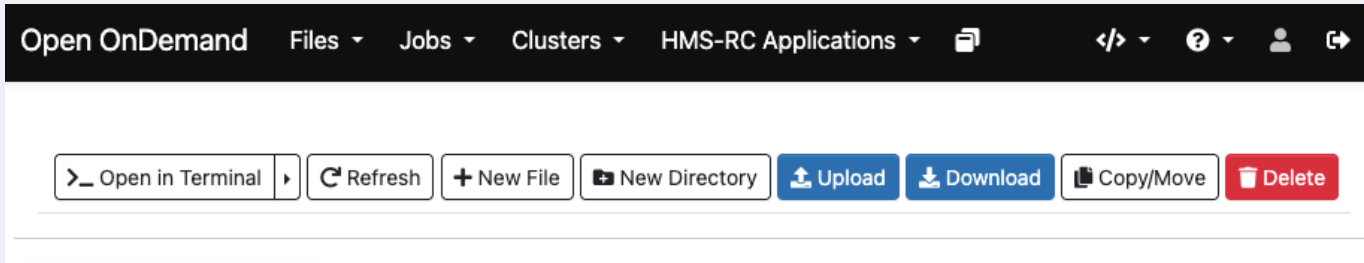
4

Rename, delete, download

From the same row of buttons.

The path bar at the top is editable — paste a long path like /n/scratch/users/m/mfk8 straight into it instead of clicking down the tree.

Getting data on and off O2



Upload

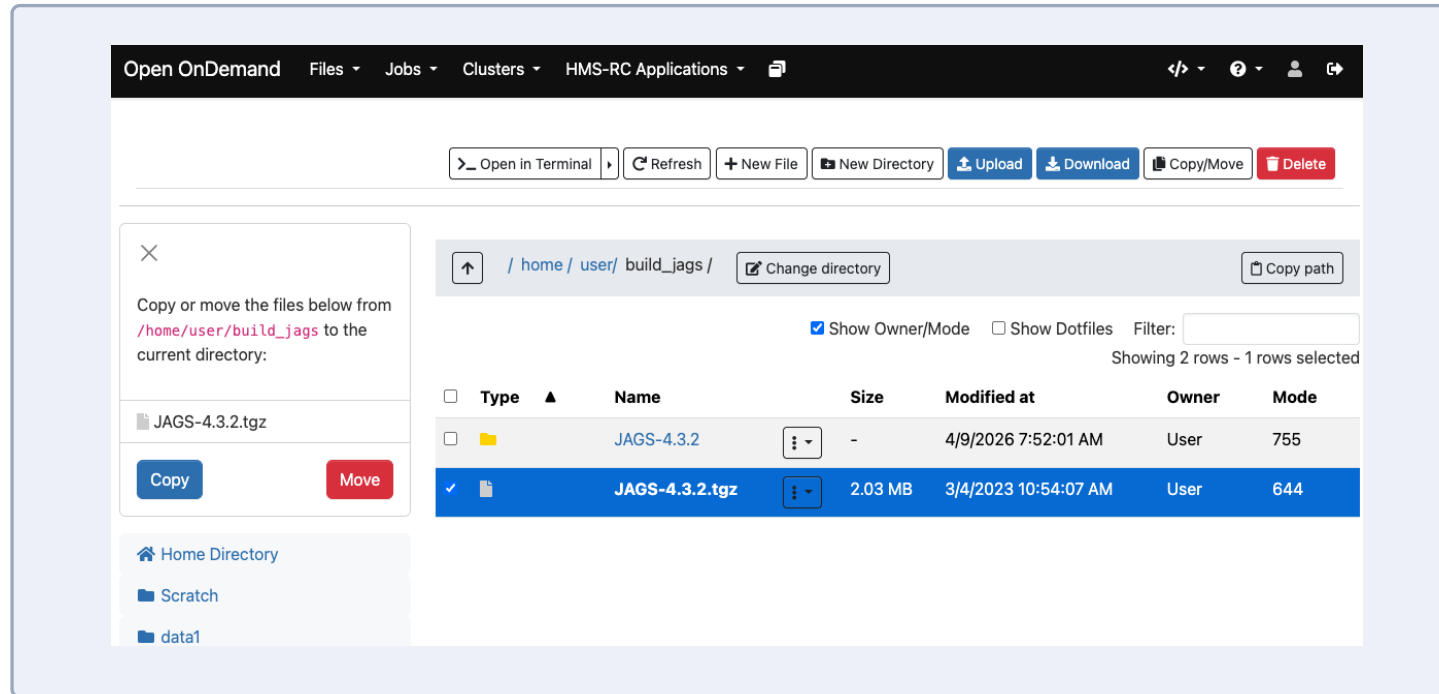
Navigate to the destination folder first, then Upload or drag files straight from your desktop into the upload browser window.

Download

Tick a file or folder and click Download. Folders arrive as a zip archive.

For very large or many-file transfers, use a purpose-built tool. Globus for large or resumable transfers, rsync or scp over the transfer cluster, or an FTP client to transfer.rc.hms.harvard.edu. Browser uploads are best for tens of files, not tens of thousands.

Copying and moving data within O2



- 1 Tick the file or folder you want
- 2 Click “Copy/Move”
- 3 Navigate to the target location — including a different filesystem
- 4 Click on Copy or Move

Your clipboard survives navigation. That is what makes this useful: copy from a group folder, switch to your scratch directory, paste. Same operation as cp and mv, without the typing.

EXERCISE 1

Your first five minutes in the Portal

1

Open the Files browser

Files › Home Directory.

2

Create a folder called o2class

Use the New Directory button.

3

Upload any small file into it

Drag it in from your desktop.

4

Create a new text file and edit it

New File, then Edit. Type a line and save.

5

Find your scratch directory

Paste `/n/scratch/users/<first char>/<your id>` into the path bar. If it does not exist, that is the create-directory script from earlier.

4 minutes · raise a hand if you get stuck

RStudio, Jupyter and MATLAB — on cluster hardware

Every Portal app you launch is a real Slurm job.

1

It looks local, it runs remote

The window is in your browser; the computation is on a compute node with as much memory, CPU or GPU as you asked for.

2

It waits in the queue like anything else

The app becomes available once Slurm dispatches your job. Bigger requests wait longer.

3

It keeps running when you disconnect

Closing the tab does not kill it. It ends when the wall-time runs out or you delete it.

4

It is billed like any other job

If your lab pays for O2, Portal apps appear on the quarterly bill — so do not leave a 3-day session idle.

Available now: RStudio · Jupyter · MATLAB · Visual Studio Code · RELION · Shiny · Desktop Mate · Interactive Job Shell

Launching one: you are filling in a job request

The screenshot shows the 'Apps stack' interface with a sidebar menu on the left and a 'Jupyter form' on the right. In the sidebar, under the 'IDE' section, 'Jupyter' is highlighted with a red circle. A red line connects this circle to the 'Jupyter form' on the right. The form contains the following fields:

- Jupyter**: This app will launch a Jupyter Notebook or Jupyter Lab on one or more cores.
- Modules to be preloaded**: A text input field containing 'gcc/14.2.0 python/3.13.1'.
- Slurm Account (This is not your username)**: A dropdown menu with 'rccg' selected.
- Partition**: A dropdown menu with 'short' selected.
- WallTime requested in hours**: A text input field containing '1'.

The form asks for exactly four things

Partition

Which queue — usually interactive or short

Number of hours

Your wall-time. Ask for a bit more than you need.

Cores and memory

Right-size it: bigger requests pend longer.

GPU / extra options

Only if the app needs one.

Click Launch, then wait for the queue. When the card turns green, click Connect.

Recognise those four? They are the same four decisions as a written job script — which is section 4.

Watching — and stopping — your session

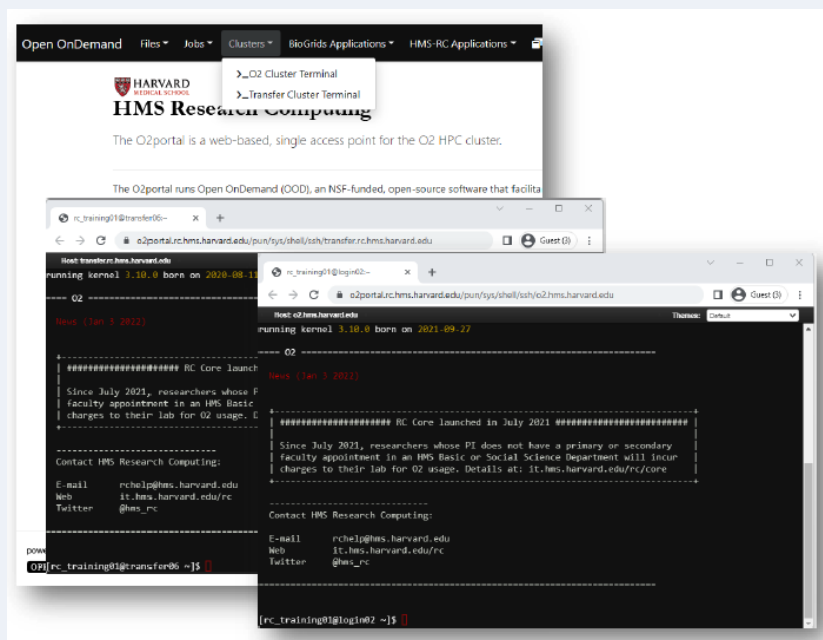
The screenshot displays the RStudio session management interface. It shows three sessions, each with a status label and a 'Delete' button. A red box labeled 'Job Status' has arrows pointing to the 'Queued', 'Running', and 'Completed' status labels.

- Queued:** RStudio Environment (59163745). Created at: 2022-08-08 17:03:32 EDT. Time Requested: 1 hour.
- Running:** RStudio Environment (14087611). 1 node | 1 core. Host: compute-e-16-230.o2.rc.hms.harvard.edu. Created at: 2023-08-01 16:05:01 EDT. Time Remaining: 58 minutes. Session ID: d262e7ff-eee2-4bb9-97e3-89c6ce0e4959. Progress & Error log: [output.log](#). Connect to RStudio Server.
- Completed:** Desktop Mate (16328564). Created at: 2025-09-22 18:42:00 EDT. Session ID: 228bf9bf-143e-422a-b4aa-f6ed09cd9e52. Progress & Error log: [output.log](#). For debugging purposes, this card will be retained for 0 more days.

- 1 My Interactive Sessions**
Every app you have running, its state, and how much time is left.
- 2 Queued → Running → Completed**
Queued means Slurm has not dispatched you yet. Be patient with large requests.
- 3 The log file link**
First place to look when an app will not start. It is the job's output.log file.
- 4 Delete ends the job**
Closing the browser tab does not. Delete when you are finished — it frees resources and stops the clock.

Delete your session when you finish — closing the tab does not. An idle 3-day RStudio session bills for 3 days.

A full O2 shell in your browser



Clusters › O2 Shell Access

Same shell as ssh, no client and no VPN. Useful when you need something the GUI does not expose:

- modules and software not wrapped as an app
- rsync, rclone or Globus command-line transfers
- copying from research.files on the transfer partition
- submitting and inspecting jobs by hand

The shell drops you on a login node. Same rule as always: start an interactive job before doing any real work — `srun --pty -p interactive -t 0-2:00 --mem 1G bash`

EXERCISE 2

Run something on a compute node

1

Pick an app from the HMS-RC menu

RStudio or Jupyter is the easiest start.

2

Request modest resources

Partition interactive, 1 hour, 1 core, 1 GB. Small requests start fastest.

3

Launch it, and watch the queue

The card moves from Queued to Running.

4

Connect, and run one line

Anything at all — you are now computing on O2.

5

Open My Interactive Sessions and find the log file

Then Delete the session when you are done.

5 minutes · small requests start fastest — do not ask for 20 cores

4

Jobs and the scheduler

What the Portal was doing on your behalf

Four decisions, every single job

A job is a request for resources, plus the commands to run once you have them. Slurm queues it, finds a node that fits, and runs it.

-p**Partition**

Which queue. Start with short; use interactive for hands-on work.

-t**Wall-time**

days-hours:minutes. Ask for ~125% of what you expect. Too long delays dispatch; too short kills the job.

-c**Cores**

Up to 20 on most partitions. Only helps if your software is actually parallel.

--mem**Memory**

Default is 1 GiB. Always in G (e.g., --mem 1G) — a bare number means MiB.

Get these four right and most job problems disappear. Everything else is refinement.

Partitions: the queues you can use

Partition	Max time	Min time	Max cores	Notes
interactive	12 hours	—	20	2 jobs at once. Defaults to 1 GiB memory.
short	12 hours	—	20	The right default for most batch jobs.
medium	5 days	12 hours	20	Will reject a job shorter than the minimum.
long	30 days	5 days	20	Will reject a job shorter than the minimum.
priority	30 days	—	20	2 running jobs at once.
gpu · gpu_quad · gpu_requeue	5 days	—	20	Extra GPU limits apply. Most GPUs are on-Quad only.
highmem	5 days	—	16	For jobs over 200 GiB. Invite-only — email rchelp.
mpi · gpu_mpi_quad	5 days	—	640 · 240	Multi-node parallel work. Invite-only.
transfer	5 days	—	4	For /n/files access. Invite-only.

Unless noted, the memory ceiling is 250 GiB per job. · Every partition can run interactive sessions, but the interactive partition has dedicated nodes and higher priority.

Which partition should I use?

Work down the list. The first yes wins.

Do you need data from /n/files (research.files)?	→	<code>transfer partition, or the transfer cluster</code>
Do you need more than 200 GiB of memory?	→	<code>highmem</code>
Do you need a GPU?	→	<code>gpu</code> · <code>gpu_quad</code> · <code>gpu_requeue</code>
Is it an MPI / multi-node job?	→	<code>mpi</code>
Are you working hands-on, one or two sessions?	→	<code>interactive</code>
Do you need one or two jobs to jump the queue?	→	<code>priority</code>
Will it finish inside 12 hours?	→	<code>short</code>
Will it finish inside 5 days?	→	<code>medium</code> – otherwise <code>long</code>

The same logic as a flowchart on the [“How to choose a partition in O2”](#) wiki page. If you are not sure, short is almost always the right first answer.

A job script, line by line

```
#!/bin/bash
#SBATCH -p short
#SBATCH -t 0-01:00
#SBATCH -c 1
#SBATCH --mem=4G
#SBATCH -o %j.out
#SBATCH -e %j.err

module load gcc/14.2.0 star/2.7.11b
STAR --genomeDir ... --readFilesIn ...
```

Submit it — and check it started

```
$ sbatch myjob.slurm
```

-p / -t / -c / --mem

The same four decisions, written down.

-o and -e

Where output and errors go. %j inserts the job ID, so runs do not overwrite each other.

module load

Make your software available — inside the job, not before it.

Then your commands

Exactly what you would type by hand.

Write a script rather than using --wrap. It is reproducible.

Watching your jobs, and loading software

Jobs

```
$ O2squeue
    what's running or pending now
$ O2_jobs_report
    past jobs + how efficient they were
$ scancel <jobid>
    stop a job
$ sacct · queue
    the raw Slurm equivalents
```

Software: Lmod modules

```
$ module load gcc/14.2.0
    load the compiler first
$ module avail
$ module spider <name>
    find what's installed
$ module load R/4.4.2
$ module list · module purge
    see loaded · clear everything
```

Also available, without you installing anything: R, Python and Conda, Perl, MATLAB · Singularity/Apptainer containers, including shared nf-core pipelines at /n/app/containers/shared · public reference databases under /n/shared_db. Need something else? Email rchelp@hms.harvard.edu — do not fight it alone.

So which do I use?

Reach for the Portal

- Interactive analysis in R, Python or MATLAB
- Anything with a graphical interface
- Moving a handful of files
- Editing a script quickly
- Checking on jobs and reading logs
- You are new to HPC with a tight deadline

Reach for the terminal

- Submitting many jobs, or job arrays
- Scripted and reproducible pipelines
- Large transfers with rsync, rclone or Globus
- Anything you want to automate
- Software installs and Conda environments

Most people end up using both — and the Portal has a terminal in it, so you never have to choose in advance.

The commands worth remembering

Getting a compute node

<code>srun --pty -p interactive -t 0-2:00 --mem 4G bash</code>	start an interactive session
<code>sbatch myjob.slurm</code>	submit a batch job
<code>scancel <jobid></code>	cancel it

Finding your way around

<code>ls</code> · <code>ls -l</code>	list a folder
<code>cd <dir></code> · <code>cd ..</code>	move in · move up
<code>less <file></code> · <code>q</code>	read a file · quit
<code>cp -r</code> · <code>mv</code> · <code>rm -r</code>	copy · move · delete
<code>*.fastq</code>	wildcard: all matching files

Checking on things

<code>O2squeue</code>	my current jobs
<code>O2_jobs_report</code>	past jobs and efficiency
<code>quota-v2</code>	my storage usage
<code>module avail</code> · <code>module load</code>	find · load software

Full list: “O2 Command Cheatsheet” on the O2 wiki. And *rm* has no *undo** — there is no trash on O2.

Where to get help

Ask us

rhelp@hms.harvard.edu

Office hours — Wednesdays, 1:00 - 3:00 pm

rc.hms.harvard.edu/office-hours

HMS IT Service Desk — (617) 432-2000

Read the wiki

The O2 space in Confluence is the source of truth for everything in this class:

Orientation for New O2 Cluster Users

O2 Cluster Status — outages and maintenance

Frequently Asked Questions

Keep learning

- 1 Intermediate O2**
Job arrays, parallel workflows, right-sizing jobs
- 2 Data management**
datamanagement.hms.harvard.edu — planning, sharing, retention
- 3 Class schedule and slides**
User Training page on the O2 wiki · github.com/hmsrc/user-training

Please fill in the course survey — it is how these sessions get better.

Questions?

rhelp@hms.harvard.edu

Office hours every Wednesday, 1:00–3:00 pm